

END SEMESTER EXAMINATIONS, SPRING 2025

Program: BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE & ENGINEERING)	
Subject Name: CRYPTOGRAPHY AND NETWORK SECURITY	
Subject Code: 2000015104	Semester: VI
Duration: 2 Hours	Maximum Marks: 50

Note: Answer the following questions as directed.

Mod. No.- Module/Topic/Unit No., **BL-** Blooms Taxonomy, **CO-** Course Outcome.

Group A

(Answer All questions)

(Each question carries 1 mark)

1. Write the correct answer from the options given: (10 X 1)

		Mod. No.	BL	CO
(i)	A virus that can be detected using antivirus software is	1	1	1
	(a) parasitic (b) worm (c) polymorphic (d) stealth			
(ii)	The _____ attack is related to confidentiality	1	1	1
	(a) interception (b) fabrication (c) Modification (d) Interruption			
(iii)	Which is among the following is NOT implemented by Digital Signature	2	2	2
	(a) Privacy (b) Authentication (c) Integrity (d) Non-repudiation			
(iv)	Which is not used in Asymmetric key Cryptography	1	1	1
	(a) RSA (b) Diffie Hellman Algorithm (c) CBC (d) ECB			
(v)	Which one of the following RC4 algorithm not used in	4	2	4
	(a) SSL (b) TLS (c) FTP (d) WEP			

		Mod. No.	BL	CO
(vi)	Which encryption algorithm uses both a public and a private key?	3	1,2	2
	(a) RSA (b) DES (c) AES (d) RC5			
(vii)	What is the primary purpose of a digital signature?	4	1	1
	(a) Data encryption (b) Data authentication (c) Data integrity (d) Data compression			
(viii)	How many rounds are presents in each iteration in SHA-3	4	2	3
	(a) 3 (b) 4 (c) 5 (d) 6			
(ix)	The 4x4 bytes in AES is called	3	1	2
	(a) States (b) Words (c) Transition (d) Permutations			
(x)	In which attack there is no modifications in the message contents	1	1	1
	(a) Active (b) passive (c) Both (a) & (b) (d) None			

Group B

(Answer any Four questions out of Six)
(Each question carries 5 marks)

(4 X 5)

2. (a) What is the Caesar cipher and how does it work?

(b) Demonstrate the encryption process for the plain text "NEIL DEGRASSE TYSON" using a shift of 3 positions. (2 + 3)

3. Describe the IDEA encryption algorithm and explain how it works.

4. Explains Advanced Encryption Standard (AES) – 128 for fixed block size of 128 bits.

Mod. No.	BL	CO
3	1	2
3	1,2	2
3	2,5	3
3	2,5	4

5. How would you define a firewall, and what are its inherent limitations in providing comprehensive network security? (1 + 4)

6. Encrypt the plaintext "*SCIENCE IS NOTHING BUT PERCEPTION*" using the PLAYFAIR cipher with the keyword "PLAYFAIREXAMPLE". Provide the cipher text along with the steps involved in the encryption process.

7. (a) Explain the concept of a hash function in cryptography. Provide examples of cryptographic hash functions.

(b) Explain Message Digest 5 (MD5) with proper flowchart. (2 + 3)

Mod. No.	BL	CO
3	2,4	3
2	4,6	3
5	1,3	2
5	1,3	2

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

8. (a) Define the term "malware" and distinguish between worms and viruses, highlighting their respective characteristics and modes of operation.

(b) Discuss different types of attacks. (5 + 5)

9. (a) Provide an overview of the RSA encryption algorithm, detailing its components and illustrating its operation with a practical example.

(b) Encrypt the plaintext "*THERE ARE NO SHORTCUTS IN EVOLUTION*" using Rail Fence Transposition Technique. (7 + 3)

Mod. No.	BL	CO
1	1,4	1
1	1,2	1
6	1,2	4
2	5	1

10. Write brief notes on any two of the following topics:

- (a) Ransomware
- (b) Firewall
- (c) Public Key Infrastructure (PKI)
- (d) TLS protocol

(5 X 2)

11. (a) Take input of two prime numbers, 13 and 19 as keys. State how the keys can be shared secretly for exchanging data over a public network through the Diffie-Hellman Key Exchange algorithm.

- (b) Differentiate between virus, worm and Trojan Horse.

(7 + 3)

Mod. No.	BL	CO
1,3,6	1,2	3
3	2,6	4
1	2	1